MECE E3610 Materials and Processes in Manufacturing
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Midterm Project Report

Group 4

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Abstract

The objective of this project was to design and fabricate a model of an early 20th-century race car using manufacturing processes studied in class: machining, silicone casting, and injection molding. Our group selected the Maserati Tipo V4 for its sleek design aesthetic and spoked wheels. Using SOLIDWORKS, we designed the main components: car body, cockpit, axles, and wheels. A two-part mold for the car body was modeled and 3D printed in PLA, then used to produce the body through silicone casting. An aluminum wheel mold was designed using CAD and machined using CAM on the Tormach. Injection molding was then used to manufacture the four wheels. Finally, all components were assembled into the completed car model. The project demonstrated a full manufacturing process from design to assembly, highlighting the importance of certain design considerations.

1 Introduction

The objective of this project was to design and manufacture a scaled model of an early 20th-century race car using manufacturing techniques introduced throughout the course. The project combined computer-aided design, additive manufacturing, machining, casting, and injection molding into a complete fabrication workflow. Our team selected the Maserati Tipo V4 as the basis for the project because of it's unique contour and aesthetic.

Task	Description
1	CAD modeling of chassis mold and cockpit
2	3D printing of chassis mold and cockpit
3	CAD modeling of wheel mold and CAM generation
4	CNC machining of wheel mold
5	Silicone casting of car body
6	Injection molding, assembly, and report preparation

Table 1: Summary of project tasks

The project was divided into several stages. First, research was conducted on the dimensions and geometry of the Maserati Tipo V4 in order to create an accurate CAD model. Using SOLIDWORKS, the assembly was separated into four primary components: the car body, cockpit, axles, and wheels. The body geometry was designed to satisfy mold and casting constraints while still maintaining the characteristics of the original car.

After the CAD phase, a two-part mold for the car body was created using negative cavity geometry. The mold was designed with alignment holes, a sprue, vents, axle holders, and a spacer. The mold halves were then fabricated using FDM 3D printing in PLA filament. Separately, a wheel mold was designed and machined from aluminum using CAM and CNC machining. Injection molding was then used to produce the wheels from the machined mold. Finally, all manufactured components were assembled to produce the completed race car model.

2 Car Body

2.1 Design

The car body was modeled in SOLIDWORKS based on reference dimensions and photographs of the Maserati Tipo V4. The design process focused on recreating the smooth shape of the vehicle while ensuring that it remained manufacturable within the constraints provided. The main feature, was to maintain the rims on the side of the car to give it that unique look. The body was scaled to maximize the allowable mold dimensions while preserving the proportions of the original vehicle.

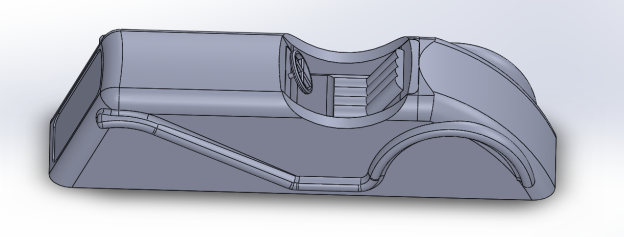


Figure 1: CAD model of the Maserati Tipo V4 car body

2.2 Mold

A two-part mold was designed for silicone casting of the car body. The mold cavity was generated by subtracting the car geometry from a rectangular mold block using the combine feature in SOLIDWORKS. Alignment holes were incorporated into both halves of the mold to ensure accurate positioning during assembly.

A sprue was placed at the highest point of the mold cavity to allow silicone to be poured effectively into the mold. Vent channels were added to reduce trapped air and improve mold filling. Axle holders were placed so the silicone could mold around, allowing space for the axle to fit on later. A spacer was added to create a cavity for the cockpit.

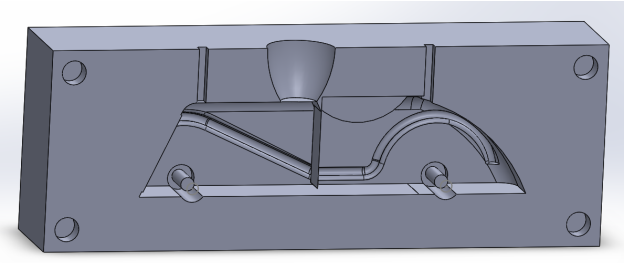


Figure 2: CAD model of the two-part car body mold

The completed mold halves were fabricated using PLA filament on a 3D printer. PLA was selected because it provided sufficient accuracy and quality for silicone casting while remaining easy to manufacture.

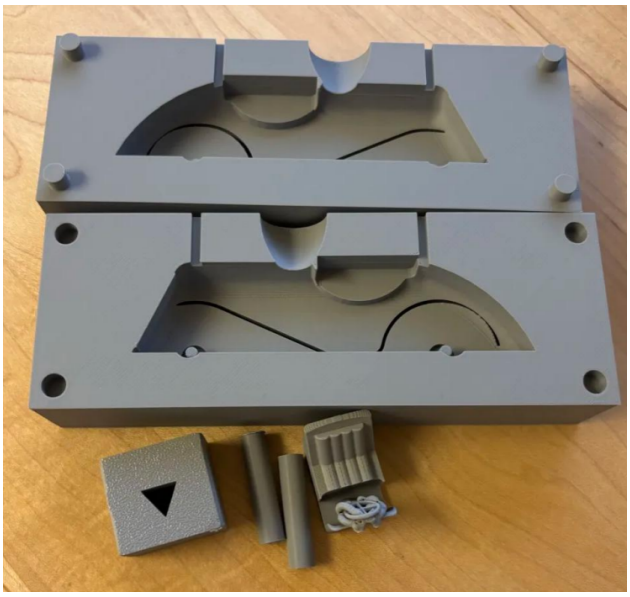


Figure 3: 3D printed mold for silicone casting

2.3 Casting

After fabrication of the mold, silicone casting was used to create the car body. The two mold halves were assembled and secured prior to pouring the silicone material through the sprue opening. The vent channels allowed trapped air to escape during filling, reducing void formation and improving casting quality.

Once the silicone cured, the mold was separated and the cast body was removed. The final silicone body preserved the overall geometry and details of the CAD model successfully. Minor texturing was present due to layer lines from the 3D printed mold, but is sufficient for our purposes.



Figure 4: Silicone casted car body

3 Wheels

3.1 Design

The wheel geometry was modeled separately in SOLIDWORKS. The design included a central hub and outer rim profile while maintaining dimensions suitable for injection molding. We incorporated spokes in our design to make the wheel look closer to the original design.

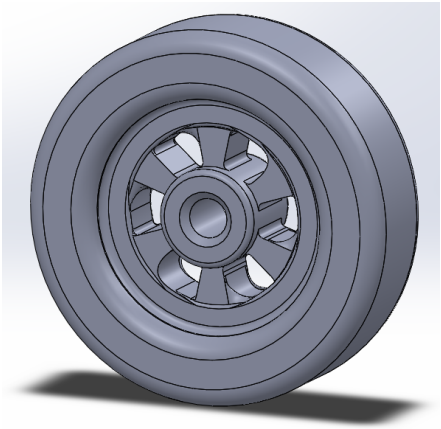


Figure 5: CAD model of the wheel

3.2 Mold

An aluminum injection mold was designed for wheel fabrication. The mold cavity was generated similar to the body using the combine feature on SOLIDWORKS. A sprue was added for the material to enter and a runner was placed at the bottom of the wheel. The mold also includes alignment holes and parallels on the side to allow the mold to be pried open after injection.

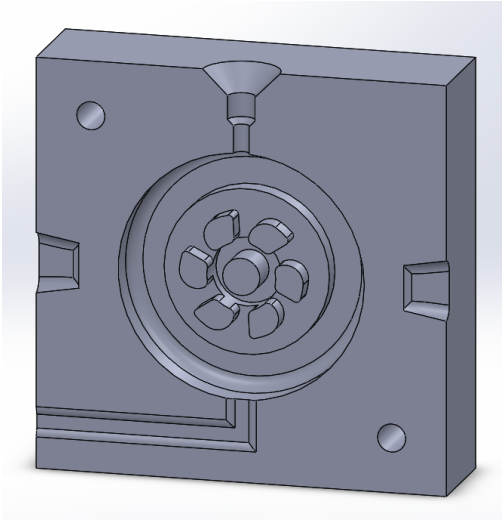


Figure 6: CAD model of the wheel injection mold

CAM software was then used to generate toolpaths and export G-code for CNC machining on the Tormach. The mold was machined from aluminum because of its durability and thermal properties, making it suitable for repeated injection molding cycles. The two molds are mirror images of each other.



Figure 7: Machined aluminum wheel mold

3.3 Injection Molding

After machining, the aluminum mold was placed in the injection molding machine. Plastic chips were placed inside the machine, melted, and then injected into the cavity to produce the wheels. Multiple wheels were fabricated using the same mold, demonstrating the repeatability of the process. The final molded wheels showed consistent geometry and dimensional accuracy. The wheels were difficult to pry out the wheels due to the spokes, causing minor deformations.

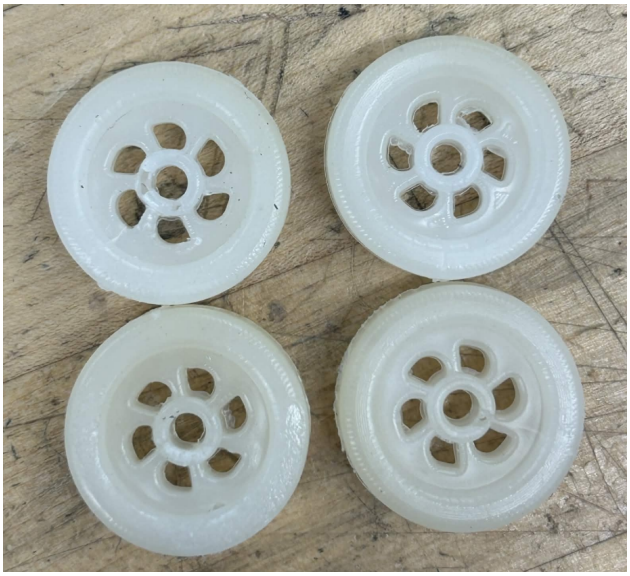


Figure 8: Injection molded wheels

4 Assembly

After all individual components were manufactured, the final assembly of the race car was completed. The silicone casted body served as the primary structure, while the separately fabricated cockpit pod was inserted into the body cavity. 3D printed rods were used as axles, and the injection molded wheels were mounted onto the axle supports. We added a 3D printed plate on the bottom of the car for extra stability and rigidity.



Figure 9: Final Assembly

The assembly process demonstrated the importance of compatibility across different man-

ufacturing processes. Because the components were designed together in CAD prior to fabrication, the final assembly fit together successfully with minimal modifications required.

5 Conclusions

In this project, a scaled model of the Maserati Tipo V4 was successfully designed and fabricated using a combination of CAD modeling, 3D printing, silicone casting, machining, and injection molding.

The use of SOLIDWORKS allowed us to have accurate digital modeling of all the components and allowed manufacturing considerations to be incorporated early on,. The silicone casting and injection molding processes both produced functional parts that were successfully integrated into the final assembly. CNC machining was accompanied to create a durable mold for the injection molding process.

Through this project, important engineering skills were put to the test from conceptual design to actual fabrication. The completed model demonstrated the effectiveness of combining multiple manufacturing techniques to produce a functional and visually accurate representation of an early race car.

References

[1] "Maserati Tipo V4," Wikipedia. [Online]. Available: https://en.wikipedia.org/wiki/Maserati_Tipo_V4

[2] S. Kalpakjian and S. Schmid, *Manufacturing Engineering and Technology*, 7th ed. Pearson, 2014.